Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Kinematics (Constant Acceleration)

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A particle accelerates from rest at $2\,\mathrm{m\,s^{-2}}$ for $5\,\mathrm{s}$. Find its final speed and the distance travelled. **[4]**

2. A car decelerates from $20\,\mathrm{m\,s^{-1}}$ to rest in $4\,\mathrm{s}$. Find the deceleration and the distance travelled. **[4]**

3. A particle moves with constant acceleration $3\,\mathrm{m\,s^{-2}}$. Its initial speed is $4\,\mathrm{m\,s^{-1}}$. Find its speed after travelling $10\,\mathrm{m}$. **[4]**

4. A ball is thrown vertically upwards at $15\,\mathrm{m\,s^{-1}}$. Find the maximum height reached (take $g = 9.8$). **[4]**

5. A particle starts from rest and accelerates at $1.5\,\mathrm{m\,s^{-2}}$ for $6\,\mathrm{s}$. Find the distance travelled. **[3]**

6. A train decelerates from $30\,\mathrm{m\,s^{-1}}$ at $2\,\mathrm{m\,s^{-2}}$. How long does it take to stop, and how far does it travel? **[4]**

7. A particle has initial velocity $5\,\mathrm{m\,s^{-1}}$ and acceleration $-2\,\mathrm{m\,s^{-2}}$. Find its velocity after $3\,\mathrm{s}$ and the distance travelled in that time. **[4]**

8. A stone is dropped from rest from a height of $45\,\mathrm{m}$. Find the time to reach the ground ($g = 9.8$). **[3]**

9. A car accelerates from $10\,\mathrm{m\,s^{-1}}$ to $25\,\mathrm{m\,s^{-1}}$ in $5\,\mathrm{s}$. Find the acceleration and the distance travelled. **[4]**

10. A particle travels 100 m in 5 s with constant acceleration, starting from rest. Find the acceleration and final speed. **[4]**

11. A ball is projected vertically upwards at $20\,\text{m s}^{-1}$. Find the time to return to the point of projection ($g = 9.8$). **[3]**

12. A particle moves from A to B with constant acceleration. It starts at $2\,\text{m s}^{-1}$ and arrives at $8\,\text{m s}^{-1}$ after travelling 30 m. Find the acceleration and time taken. **[5]**

13. A car brakes from $18\,\text{m s}^{-1}$ to $6\,\text{m s}^{-1}$ over 48 m. Find the deceleration. **[3]**

14. A particle starts with speed u and accelerates at a for time t. Write expressions for final speed and displacement. **[3]**

15. A particle is projected vertically upwards at $12\,\text{m s}^{-1}$ from a point 16 m above the ground. Find the speed when it hits the ground ($g = 9.8$). **[5]**

Section B – Hard (Q16–25)

16. A particle travels with constant acceleration. It covers 20 m in the first 2 s and 50 m in the next 3 s. Find the initial speed and acceleration. **[6]**

17. A car accelerates from rest at $2\,\text{m s}^{-2}$ for 10 s, then travels at constant speed for 20 s, then decelerates at $1\,\text{m s}^{-2}$ to rest. Find the total distance travelled. **[6]**

18. A ball is thrown upwards at $25\,\text{m s}^{-1}$ from the top of a 30 m cliff. Find the time to hit the sea and the speed on impact ($g = 9.8$). **[6]**

19. Two particles A and B start from the same point. A has constant speed $10\,\text{m s}^{-1}$. B starts from rest with acceleration $2\,\text{m s}^{-2}$. Find when and where B overtakes A. **[6]**

20. A particle moves in a straight line with constant acceleration. Its velocity is $3\,\mathrm{m\,s^{-1}}$ at $t = 0$ and $15\,\mathrm{m\,s^{-1}}$ at $t = 4$. Find the displacement during the 4th second. **[5]**

21. A train passes a station at $20\,\mathrm{m\,s^{-1}}$ and continues with constant acceleration $0.5\,\mathrm{m\,s^{-2}}$. Find how far it travels in the next 30 s. **[4]**

22. A particle is projected vertically upwards at $u\,\mathrm{m\,s^{-1}}$. Show that the time to return is $\frac{2u}{g}$ and the maximum height is $\frac{u^2}{2g}$. **[5]**

23. A car starts from rest and accelerates at $3\,\mathrm{m\,s^{-2}}$ for 4 s, then decelerates at $2\,\mathrm{m\,s^{-2}}$ until it stops. Find the total time and distance. **[6]**

24. A particle has displacement $s = 2t^2 + 3t$ (constant acceleration form). Identify u and a, and find velocity at $t = 5$. **[4]**

25. A stone is dropped from a balloon ascending at $5\,\mathrm{m\,s^{-1}}$. The stone reaches the ground after 4 s. Find the height of the balloon when the stone was dropped ($g = 9.8$). **[6]**

End of Worksheet • *Show all working clearly. Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless otherwise stated.*